

Back in the Eye of the Storm: The Visual Effects of Twisters

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ABSTRACT

We begin with an exploration of the evolution from the original 1996 *Twister* to 2024's *Twisters*, highlighting technological advancements and their impact on visual storytelling. The seamless integration of special effects (SFX) and visual effects (VFX) is crucial for achieving realism, and we will discuss the collaboration between these teams to create cohesive, immersive experiences. A key focus is on the elemental analysis of storms, breaking down the visual components that form tornadoes. This includes understanding the recipe and rules of storms, their lifecycle, and their environmental impact. The tools and setups developed to recreate these phenomena will be showcased, illustrating the blend of science and creativity. Photography plays a vital role in capturing the essence of real storms. We will examine the extensive reference photography process, from hi-res storm chaser footage driving into storms to everyday iPhone and GoPro clips, and how these references shape our imagination and design of tornadoes. Finally, the talk will cover the process of transforming tornadoes into characters within the narrative. Each tornado in *Twisters* is unique in size, shape, and form, following the simple physics of these natural phenomena while pushing creative boundaries. This section will reveal how Industrial Light & Magic (ILM) rigged and animated different tornadoes, making them central figures in the story. This talk offers an in-depth look at the magic that brings nature's fury to life, inspiring and educating the audience on the potential of visual effects in film to replicate nature.

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1 FROM TWISTER TO TWISTERS: THE PAST, PRESENT, AND FUTURE

Twister (1996), the disaster thriller film directed by Jan de Bont set a high bar for weather simulations with ILM's pioneering use of computer graphics (CG). At the time, the technology was in its early stages, and the production team had to invent new techniques to create realistic tornadoes. The film's groundbreaking visual effects

were achieved through a combination of practical effects and early CG, setting a precedent for future weather-related films. Early particle system toolsets were used and new techniques were developed for rendering particles including the ability to create shadowing. Geometry based rendering with fractal shaders was used for some of the tornadoes and roiling clouds and a lot of the look of the final images relied on compositing and image processing.

Fast forward to 2024, and Lee Isaac Chung's *Twisters* leverages the latest technology to deliver even more realistic and mesmerizing depictions of tornadoes. This section delves into the technological advancements over the years, including improvements in computational power, and simulation and rendering techniques that have transformed the VFX landscape. Innovations in digital visual effects, combined with practical special effects, have allowed filmmakers to create tornadoes that are not only more visually stunning but also more believable. The comparison between *Twister* and *Twisters* highlights the evolution of visual effects in cinema.

2 PHOTOGRAPHY OF A STORM: REFERENCE PHOTOGRAPHY

Before creating realistic tornadoes and storms, it is crucial to observe and understand what they look like in reality. The filmmakers of *Twisters* sent professional storm-chaser Sean Casey into real storms who provided original footage of supercell clouds and tornadoes, captured on professional hi-res cameras. These references became valuable visual assets for the VFX team. Additionally, ILM sent photographers into the Midwest to capture virtual backgrounds of stormy skies, ensuring a broad range of references and image assets.

By sending real storm chasers into the field and capturing detailed footage, the filmmakers ensured that every tornado in *Twisters* was grounded in reality. This approach provided the VFX team with essential visual assets and allowed for a deeper understanding of storm dynamics. The result is a film that balances scientific accuracy with creative storytelling, pushing the boundaries of what is possible in visual effects.

3 INTERSECTION AND BLEND BETWEEN SFX AND VFX

Producing the film's practical, on-set tornado effects involved generating wind, rain, hail, water and flying debris. Jet engines and wind-producing machines were used to create high-impact wind effects. These machines provided powerful gusts of wind to simulate the intense conditions of a tornado. Additionally, dump tanks and high-pressure pneumatic water cannons were used to create realistic water effects.

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The practical effects provided a foundation for the VFX team, who then augmented these elements with digital enhancements. Advanced tools and techniques allowed the VFX team to create complex and realistic CG weather simulations. The ten digital tornadoes and weather environments were designed to not only look realistic but also behave authentically, adhering to the natural laws of physics. This ensured a seamless intersection of SFX and VFX, resulting in a visually immersive experience.

4 ANALYZING AND REPLICATING NATURE: THE RECIPE AND RULES OF A STORM

Creating a believable storm requires a detailed understanding of its visual elements. The VFX team needed to unpack the components of a storm, from cloud formation to the destruction it causes. The process involves an elemental breakdown of a storm, and understanding the visual evolution, formation, and life cycle of storms. The shaping, advection, and rendering of clouds are critical elements. By adhering to the rules of a storm and not breaking physical bounds, filmmakers can create realistic and compelling tornado simulations. Getting all the size metrics right is essential to ensure the accuracy and believability of the storm sequences.

5 TORNADOES AS MONSTERS: MAKING A CHARACTER OUT OF NATURAL PHENOMENA

In *Twisters*, tornadoes are not just destructive forces but are portrayed as monstrous entities with distinct personalities. This section explores how tornadoes are conceptualized as villains, akin to how *Jaws* depicted its shark. Each tornado in the film varies in size, shape, form, danger, and visual complexity, becoming a character in its own right.

Rigging and animating different tornadoes involve creating unique behaviors and appearances for each storm. This process includes layout, creative development, animation, effects, and composition. The filmmakers followed simple physics principles to ensure the tornadoes behaved authentically, without breaking physical bounds. Each tornado has a unique look and impact on the story which helped the filmmakers enhance the film's emotional depth.

6 CONCLUSION

The process of artistically and scientifically analyzing and replicating nature is at the heart of creating realistic weather simulations in film. From the early days of *Twister* to the advanced techniques used in *Twisters*, the continuous innovation in visual effects demonstrates a dedication to authenticity and artistic expression. By studying real-world phenomena, leveraging cutting-edge technology, and creatively enhancing natural elements, filmmakers can produce visually compelling and accurate depictions of storms.